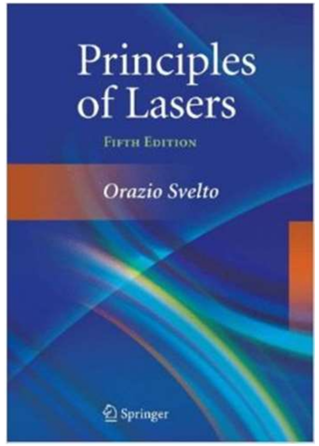
《激光原理与技术》

Principles and Technologies of Lasers



Ch2-4
Open Cavity & Gaussian Beam

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Chapter 2: Open Cavity & Gaussian Beam

❖ 2.6 The traveling-wave field of square confocal cavity

- The Hermite-Gaussian beam in confocal cavity
- Amplitude distribution & beam size
- Mode volume
- Constant phase surface
- Far field divergence angle

❖ 2.7 The circular confocal cavity

❖ 2.8 Characteristics of general stable spherical cavity

2.6 The traveling-wave field of square confocal cavity

1) The Hermite-Gaussian beam in confocal cavity

Fresnel Kirchhoff's Law

Field on the mirror $\longrightarrow$ Field at any position

$$V(x,y,z)=\frac{ik}{4\pi}\iint_{s_1}V(x_1,y_1)\frac{e^{-ik\rho}}{\rho}(1+\cos\theta)\,ds_1$$

$$V_{mn}(x,y)=C_{mn}H_m\left(\sqrt{\frac{2\pi}{L\lambda}}x\right)H_n\left(\sqrt{\frac{2\pi}{L\lambda}}y\right)e^{-\frac{x^2+y^2}{(L\lambda/\pi)}}$$

The field distribution of the mode

$$E_{mn}(x,y,z)=A_{mn}E_0\frac{w_0}{w(z)}H_m\left[\frac{\sqrt{2}}{w(z)}x\right]H_n\left[\frac{\sqrt{2}}{w(z)}y\right]e^{-\frac{r^2}{w^2(z)}}e^{-i\Phi(x,y,z)}$$

E_0 and A_{mn} are constants, $w_0=w_{0s}/\sqrt{2}=\sqrt{L\lambda/2\pi}$, w_{0s} beam size on the mirror

The electric field of TEM_{mn} on position (x,y,z)

$$E_{mn}(x,y,z)=A_{mn}E_0\boxed{\frac{w_0}{w(z)}}\boxed{H_m\left[\frac{\sqrt{2}}{w(z)}x\right]H_n\left[\frac{\sqrt{2}}{w(z)}y\right]e^{-\frac{r^2}{w^2(z)}}}\boxed{e^{-i\Phi(x,y,z)}}$$

$$\boxed{\frac{w_0}{w(z)}}$$

Amplitude Attenuation Factor: reflecting the amplitude change with propagation

$$w(z)=\sqrt{\frac{L\lambda}{2\pi}\left(1+\frac{z^2}{f^2}\right)}=\frac{w_{0s}}{\sqrt{2}}\sqrt{1+\left(\frac{z}{f}\right)^2}=w_0\sqrt{1+\left(\frac{z}{f}\right)^2}$$

$$\boxed{H_m\left[\frac{\sqrt{2}}{w(z)}x\right]H_n\left[\frac{\sqrt{2}}{w(z)}y\right]e^{-\frac{r^2}{w^2(z)}}}$$

Transverse Amplitude Distribution Factor: reflecting the distribution of the field amplitude in transverse direction

$$\boxed{e^{-i\Phi(x,y,z)}}$$

Phase Factor: determine the phase distribution inside the confocal cavity

$$\Phi(x,y,z)=k\left[\underbrace{f(1+\xi)}_{\text{Propagation factor}}+\underbrace{\frac{\xi}{1+\xi^2}\frac{r^2}{2f}}_{\text{Phase bending factor}}\right]-\underbrace{(m+n+1)\left(\frac{\pi}{2}-\psi\right)}_{\text{Additional phase shift}}$$

$$\psi=arctg\left[{(1-\xi)}\Big/{(1+\xi)}\right];\xi={2z}/{L}={z}/{f};w_0={w_{0s}}/{\sqrt{2}}=\sqrt{{L\lambda}/{2\pi}}$$

2) Amplitude distribution & beam size

The amplitude $$\left|E_{mn}(x,y,z)\right| = A_{mn}E_0\frac{w_0}{w(z)}H_m\left[\frac{\sqrt{2}}{w(z)}x\right]H_n\left[\frac{\sqrt{2}}{w(z)}y\right]e^{-\frac{x^2+y^2}{w^2(z)}}$$

For fundamental mode $$\left|E_{00}(x,y,z)\right| = A_{00}E_0\frac{w_0}{w(z)}e^{-\frac{x^2+y^2}{w^2(z)}}$$

The beam size $$w(z)=\sqrt{\frac{L\lambda}{2\pi}\left(1+\frac{z^2}{f^2}\right)}=\frac{w_{0s}}{\sqrt{2}}\sqrt{1+\left(\frac{z}{f}\right)^2}=w_0\sqrt{1+\left(\frac{z}{f}\right)^2}$$

When on the mirror $$w(z)=w(\pm f)=w_{0s}=\sqrt{\frac{L\lambda}{\pi}}$$

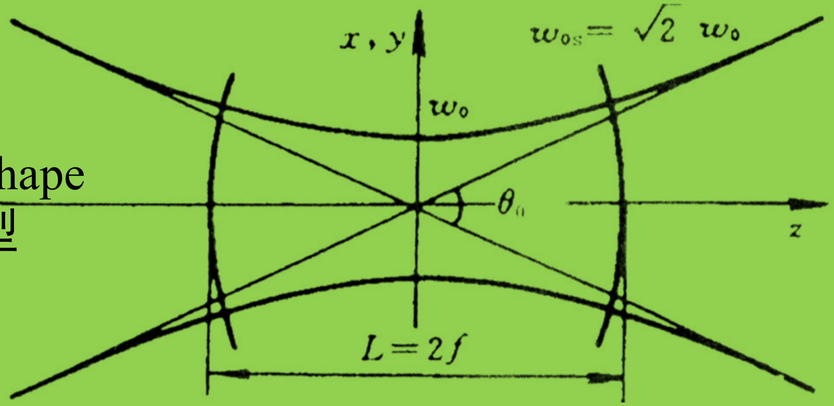
Beam waist $$w_0=w(0)=\frac{w_{0s}}{\sqrt{2}}=\sqrt{\frac{L\lambda}{2\pi}}=\sqrt{\frac{f\lambda}{\pi}}$$

基模高斯光束的束腰半径

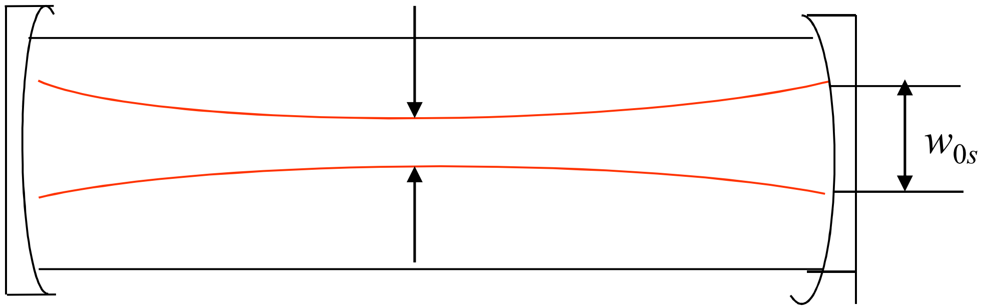
发散角和位置无关，所有位置发散角一样

$$\boxed{\frac{w^2(z)}{w_0^2}-\frac{z^2}{f^2}=1}$$

Hyperbola shape
双曲线型



3) Mode volume: Spatial expansion range of mode inside the cavity



- Mode volume → Contribution of excited state carriers → Output power
- The base mode volume of the confocal cavity

$$\boxed{V_{00}^{0}=\frac{1}{2}L\pi w_{0s}^{2}=\frac{L^{2}\lambda}{2}}$$

- For high order mode: order number ↑, mode volume ↑

$$V_{mn}^{0}=\frac{1}{2}L\pi w_{ms}w_{ns}=\sqrt{(2m+1)(2n+1)}\,\frac{L^{2}\lambda}{2}$$

Example： λ=10.6 μm, L=1 m, 2a=20 mm, V=314 cm^3 → V_{00}^{0}=5.3 cm^3

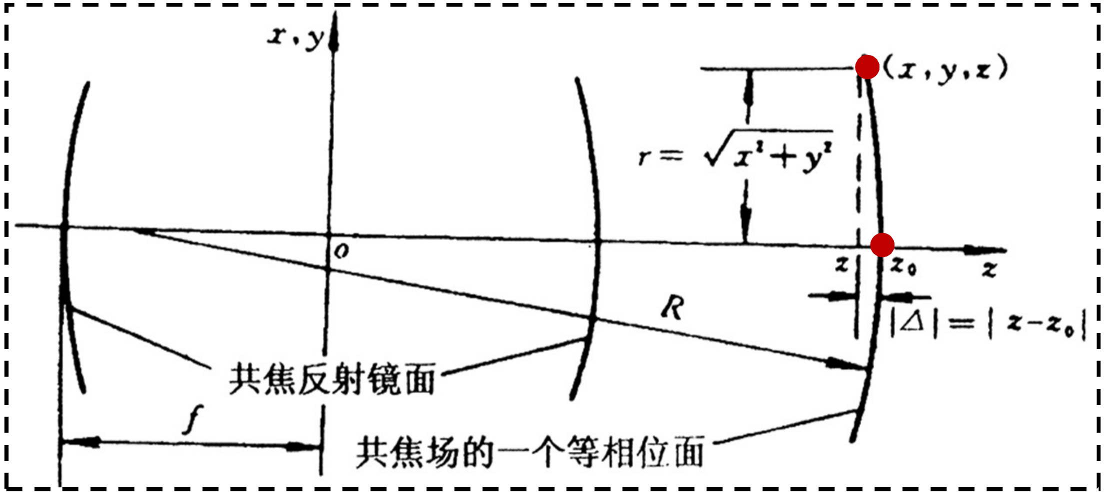
$$V_{00}^{0}/V=5.3/314=1.7\%$$

4) Equivalent phase surface (wave-front)

$$E_{mn}(x,y,z) = A_{mn} E_0 \frac{w_0}{w(z)} H_m\left[\frac{\sqrt{2}}{w(z)} x\right] H_n\left[\frac{\sqrt{2}}{w(z)} y\right] e^{-\frac{r^2}{w^2(z)}} \underline{e^{-i\Phi(x,y,z)}}$$

$$\Phi(x,y,z) = k\left[f(1+\xi) + \frac{\xi}{1+\xi^2}\frac{r^2}{2f}\right] - (m+n+1)\left(\frac{\pi}{2} - \psi\right)$$

where $\xi = \frac{2z}{L} = \frac{z}{f}$ $\quad \psi = arctg\left(\frac{1-\xi}{1+\xi}\right)$



For equivalent phase surface

$$\boxed{\Phi(x,y,z) = \Phi(0,0,z_0)}$$

$$\Phi(0,0,z_0) = k\left[f(1+\xi_0)\right] - (m+n+1)\left(\frac{\pi}{2} - \psi_0\right) \qquad \xi_0 = \frac{z_0}{f}$$

$$k\left[\underbrace{f\left(1+\xi\right)}_{=f+z}+\frac{\xi}{1+\xi^2}\frac{r^2}{2f}\right]-\cancel{\left(m+n+1\right)\left(\frac{\pi}{2}-\psi\right)}=k\left[\underbrace{f\left(1+\xi_0\right)}_{=f+z_0}\right]-\cancel{\left(m+n+1\right)\left(\frac{\pi}{2}-\psi_0\right)}$$

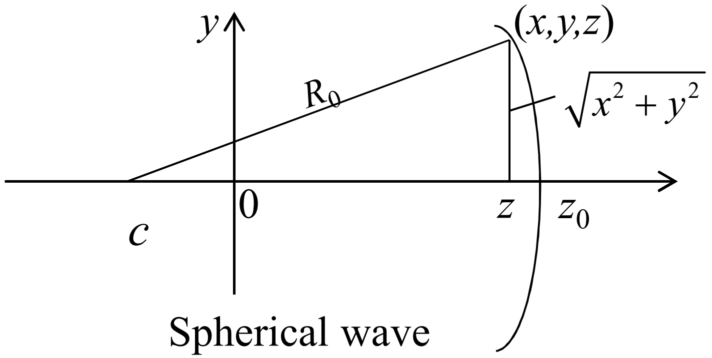
For paraxial light $\quad \psi \approx \psi_0$

$$\left|z-z_0\right|=\frac{\xi}{1+\xi^2}\frac{x^2+y^2}{2f}\approx\frac{\xi_0}{1+\xi_0{}^2}\frac{r^2}{L}=\frac{x^2+y^2}{\left(1+\xi_0{}^2\right)\dfrac{L}{\xi_0}}$$

parabolic equation
抛物面方程

Gaussian beam can be approximately treated as spherical wave near the axis?
高斯光波在腔轴附近可近似为球面波？

1. Focus point c locates at $-z$



$$x^2+y^2+\left[z+\left(R_0-z_0\right)\right]^2=R_0^2$$

$$z-z_0+R_0=\sqrt{R_0^2-\left(x^2+y^2\right)}$$

$$\boxed{\begin{array}{c} x^2+y^2<<R_0^2 \\ 1^{\text{st}}\text{ order}\end{array}} \searrow \approx R_0\left(1-\frac{x^2+y^2}{2R_0^2}\right)$$

$$z - z_0 + R_0 \approx R_0\left(1 - \frac{x^2 + y^2}{2R_0^2}\right) \quad \Rightarrow \quad z - z_0 \approx -\frac{x^2 + y^2}{2R_0}$$

Spherical wave near the axis
近轴球面波

Gaussian beam near the axis
近轴高斯光波

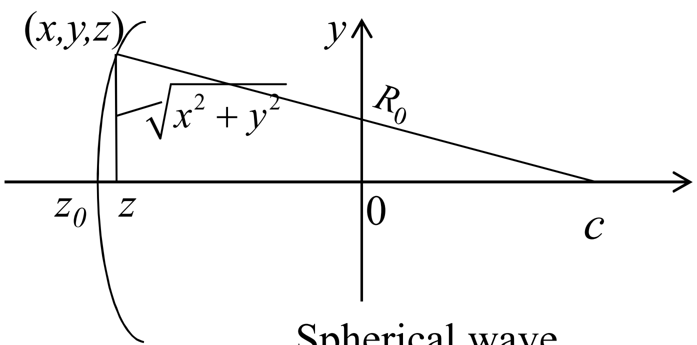
parabolic equation
抛物面方程

$$z - z_0 \approx -\frac{x^2 + y^2}{2R_0} \quad \Leftrightarrow \text{v.s.} \Leftrightarrow \quad z - z_0 = -\frac{x^2 + y^2}{\left(1 + \xi_0^{\ 2}\right)\dfrac{L}{\xi_0}}$$

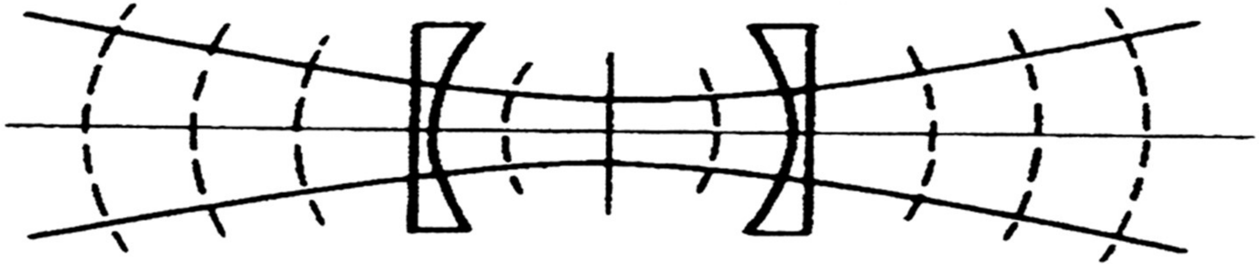
$$\Rightarrow \quad R_0 = z_0\left[1 + \left(\frac{L}{2z_0}\right)^2\right] \qquad \left(z_0 > 0\right)$$

2. Focus point c locates at $+z$, similar treatment

$$\Rightarrow \quad R_0 = -z_0\left[1 + \left(\frac{L}{2z_0}\right)^2\right] \qquad \left(z_0 < 0\right)$$



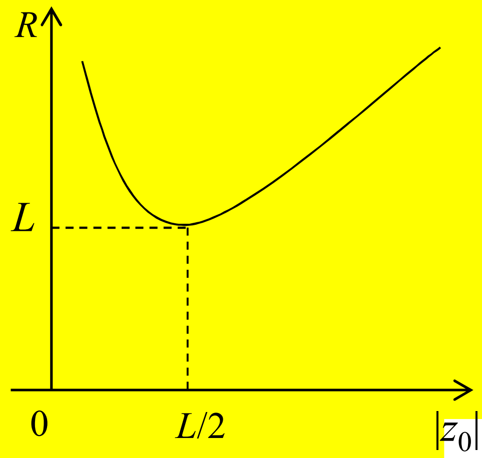
Spherical wave



$$R(z_0)=\left|z_0+\frac{f^2}{z_0}\right|=\left|f\left(\frac{z_0}{f}+\frac{f}{z_0}\right)\right|=|z_0|\left[1+\left(\frac{L}{2z_0}\right)^2\right]$$

$z_0=0 \qquad R(z_0)\to\infty$ Center of the confocal cavity
Wave-front perpendicular to the axis

$z_0\to\infty \qquad R(z_0)\to\infty$ Plane wave perpendicular to the axis

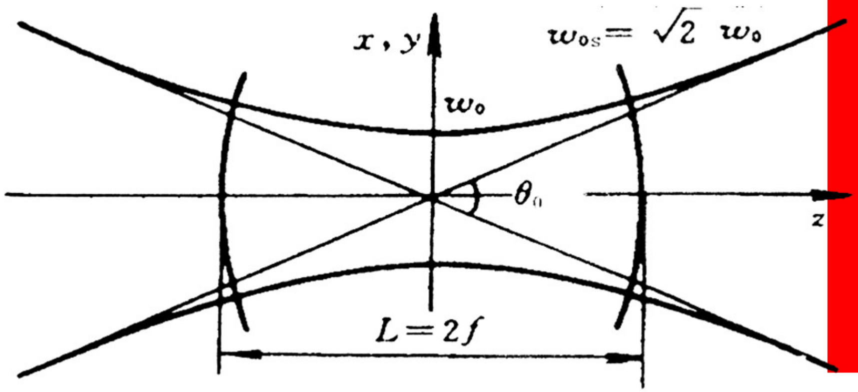


$z_0=\pm|z_0| \qquad R(z_0)$ is the same, symmetry spherical wave

$z_0=\pm f=\pm L/2 \qquad R(z_0)=2f=L$ Wave-front coincident with mirros

5) Far-field divergence angle

$$\theta = \lim_{z\to\infty}\frac{2w(z)}{z}$$



$$\theta_{1/e^2} = \lim_{z\to\infty}\frac{2\sqrt{\frac{L\lambda}{2\pi}\left(1+\frac{z^2}{f^2}\right)}}{z} = 2\sqrt{\frac{\lambda}{f\pi}}$$

He-Ne : L=30 *cm* λ=632.8 *nm* $\theta_{1/e^2} \sim 2.3\times 10^{-3}$ rad

CO_2 : L=100 *cm* λ=10.6 *mm* $\theta_{1/e^2} \sim 5.2\times 10^{-3}$ rad

- The divergence angle is small, only 10^{-3} rad.
- The divergence angle is bigger for higher order mode

$$\theta_m = \sqrt{2m+1}\,\theta_0 \qquad \theta_n = \sqrt{2n+1}\,\theta_0$$

2.7 The Circular Confocal Cavity

Laguerre-Gaussian approximate

Eigen Function $$V_{mn}(r,\phi)=C_{mn}\left(\sqrt{2}\frac{r}{w_{0s}}\right)^m L_n^m\left(2\frac{r^2}{w_{0s}{}^2}\right)e^{-\frac{r^2}{w_{0s}^2}}e^{-im\phi}$$

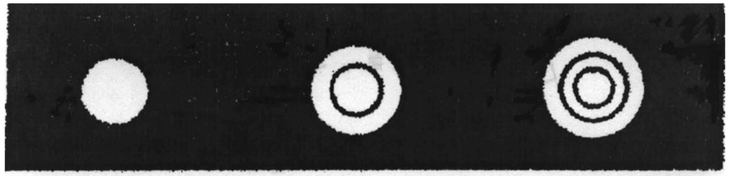
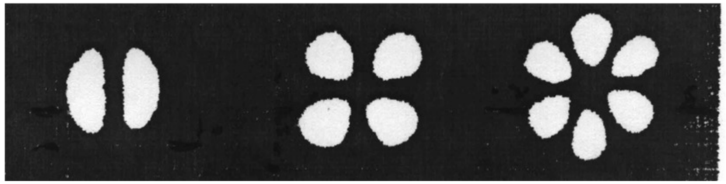
Eigen Value $$\gamma_{mn}=e^{i\left[kL-(m+2n+1)\frac{\pi}{2}\right]}$$

Laguerre Polynomial $L_0^m(\varsigma)=1$ $\quad L_1^m(\varsigma)=1+m+\varsigma$ $\quad\ldots$

Fundamental mode $$V_{00}(r,\phi)=C_{00}e^{-r^2/w_{0s}^2}$$

High-order mode
$$\begin{cases} V_{10}(r,\phi)=C_{10}\dfrac{\sqrt{2}}{w_{0s}}re^{-\frac{r^2}{w_{0s}^2}}e^{-i\phi} \\ V_{01}(r,\varphi)=C_{01}\left(1-2\dfrac{r^2}{w_{0s}^2}\right)e^{-\frac{r^2}{w_{0s}^2}} \end{cases}$$

- The amplitude distribution of the mode

TEM$_{00}$ TEM$_{01}$ TEM$_{02}$ TEM$_{10}$ TEM$_{20}$ TEM$_{30}$

- Phase shift $\delta\varphi_{mn} = -kL + (m+2n+1)\dfrac{\pi}{2} = -kL + \Delta\varphi_{mn}$

- Resonant frequency $\nu_{mnq} = \dfrac{c}{2L'}\left[q + \dfrac{1}{2}(m+2n+1)\right]$

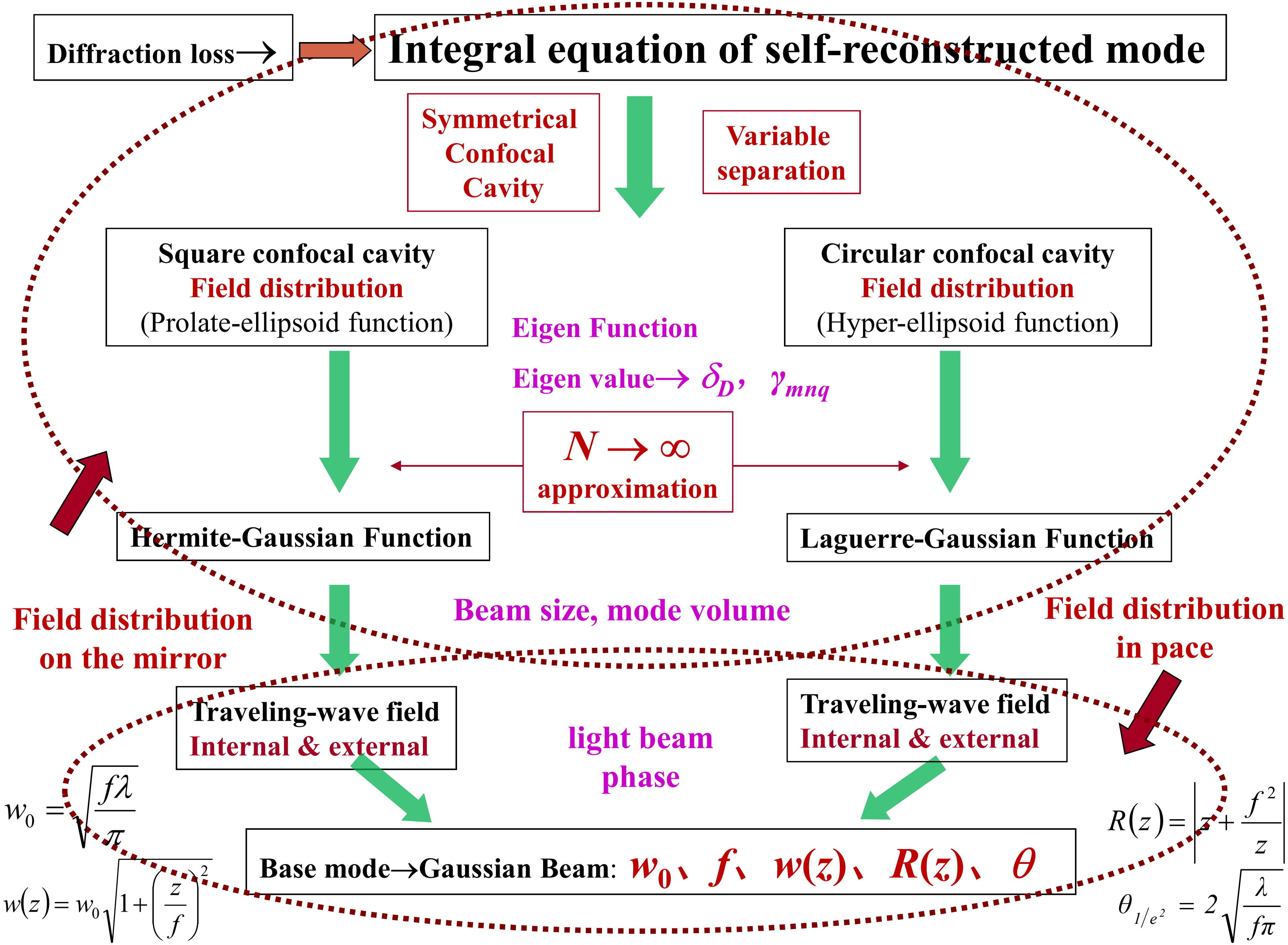
- Diffraction loss $\delta_{mn} = 1 - \left|\dfrac{1}{\gamma_{mn}}\right|^2$

- The traveling-wave field of circular confocal cavity

$$E_{mn}(r,\phi,z) = A_{mn}E_0\frac{w_0}{w(z)}\left(\sqrt{2}\frac{r}{w(z)}\right)^m L_n^m\left(2\frac{r^2}{w^2(z)}\right)\mathrm{e}^{-\frac{r^2}{\omega^2(z)}}\mathrm{e}^{-\mathrm{i}m\phi}\mathrm{e}^{-\mathrm{i}\Phi(r,\phi,z)}$$

Type of Confocal Cavity	Analytical solution	Approximation
Square	Prolate-ellipsoid function	Hermite-Gaussian function
Circular	Super-ellipsoid function	Laguerre-Gaussian function

- The beam characteristics of confocal cavity only determine by focal length f, not related to the size of the mirror. Parameters f or w_0 determines the Gaussian beam inside the cavity.
- Analytical solution represents the real case of loss inside cavity. The loss of the transverse mode is determine by the Fresnel number. Different transverse modes have different losses.
- Characteristics of confocal cavity:
  - ✓ The diffraction loss is low.
  - ✓ Mode degeneracy.
  - ✓ Light spot size along the cavity axis follows the hyperbolic function.
  - ✓ Equivalent phase surface (spherical).
  - ✓ The phase surface coincide with the mirror.

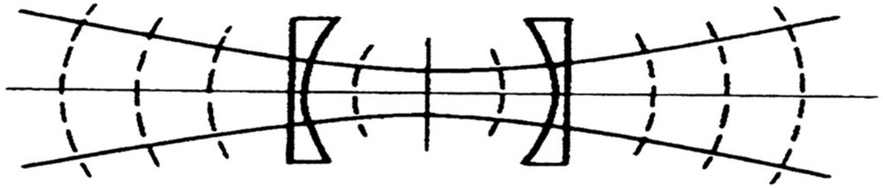

Diffraction loss→
Integral equation of self-reconstructed mode
Symmetrical Confocal Cavity
Variable separation
Square confocal cavity
Field distribution
(Prolate-ellipsoid function)
Circular confocal cavity
Field distribution
(Hyper-ellipsoid function)
Eigen Function
Eigen value→ δ_D, γ_{mnq}
$N \to \infty$
approximation
Hermite-Gaussian Function
Laguerre-Gaussian Function
Field distribution on the mirror
Beam size, mode volume
Field distribution in pace
Traveling-wave field
Internal & external
light beam phase
Traveling-wave field
Internal & external
$w_0 = \sqrt{\frac{f\lambda}{\pi}}$
$w(z) = w_0\sqrt{1+\left(\frac{z}{f}\right)^2}$
Base mode→Gaussian Beam: w_0、f、$w(z)$、$R(z)$、θ
$R(z) = \left|z + \frac{f^2}{z}\right|$
$\theta_{1/e^2} = 2\sqrt{\frac{\lambda}{f\pi}}$

2.8 Characteristics of general stable spherical cavity

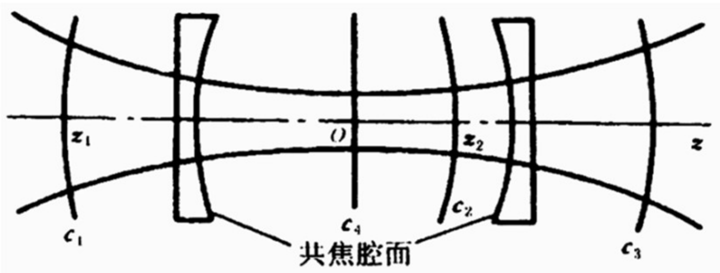
Basic idea: confocal cavity = = general stable spherical cavity

Any confocal cavity can be equivalent to an infinite number of stable spherical cavities, while any stable spherical cavity can only have one equivalent confocal cavity

Equivalent: have the same traveling wave



Any confocal cavity can be equivalent to an infinite number of stable spherical cavities

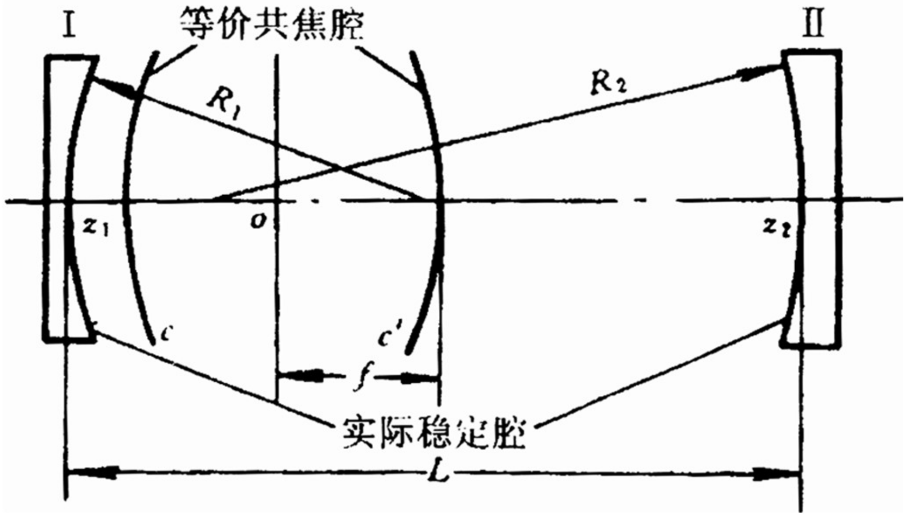


$$\begin{cases} R_1 = R(z_1) = -\left(z_1 + \dfrac{f^2}{z_1}\right) \\ R_2 = R(z_2) = +\left(z_2 + \dfrac{f^2}{z_2}\right) \\ L = z_2 - z_1 \end{cases}$$

$$0 < \left(1 - \frac{L}{R_1}\right)\left(1 - \frac{L}{R_2}\right) < 1$$ is stable spherical cavity

Any stable spherical cavity can only have one equivalent confocal cavity

Problem: know R_1, R_2, L, how to determine the position and f of confocal cavity



$$
\begin{cases}
R_1 = R(z_1) = -\left(z_1 + \dfrac{f^2}{z_1}\right) \\
R_2 = R(z_2) = +\left(z_2 + \dfrac{f^2}{z_2}\right) \\
L = z_2 - z_1
\end{cases}
$$

$$z_1 = \frac{L(R_2 - L)}{(L - R_1) + (L - R_2)} \qquad z_2 = \frac{-L(R_1 - L)}{(L - R_1) + (L - R_2)}$$

$$f^2 = \frac{L(R_1 - L)(R_2 - L)(R_1 + R_2 - L)}{\left[(L - R_1) + (L - R_2)\right]^2} \qquad \text{if} \quad 0 < (1 - L/R_1)(1 - L/R_2) < 1$$

We have $f^2 > 0, z_1 < 0, z_2 > 0$

Mode characteristics of general stable spherical cavity

1. Beam spot size on the mirror (base mode)

$$\left|E_{00}(x,y,z)\right| = A_{00}E_0 \frac{w_0}{w(z)} e^{-\frac{r^2}{w^2(z)}}$$

$$w(z) = w_0\sqrt{1+\left(\frac{z}{f}\right)^2}$$

Insert the expression of z_1, z_2 and f

$$\left.\begin{aligned} w_{s1} &= \sqrt{\frac{\lambda L}{\pi}}\left[\frac{R_1^2(R_2-L)}{L(R_1-L)(R_1+R_2-L)}\right]^{1/4} \\ w_{s2} &= \sqrt{\frac{\lambda L}{\pi}}\left[\frac{R_2^2(R_1-L)}{L(R_2-L)(R_1+R_2-L)}\right]^{1/4} \end{aligned}\right\}(2.8.6) \text{ or } \left.\begin{aligned} w_{s1} &= w_{0s}\times\left[\frac{g_2}{g_1(1-g_1g_2)}\right]^{1/4} \\ w_{s2} &= w_{0s}\times\left[\frac{g_1}{g_2(1-g_1g_2)}\right]^{1/4} \\ w_{0s} &= \sqrt{L\lambda/\pi} \end{aligned}\right\}(2.8.7)$$

only for stable cavity

2. Mode volume (base mode)

$$\left.\begin{aligned} V_{00} &= \frac{1}{2}L\pi\left(\frac{w_{s1}+w_{s2}}{2}\right)^2 = V_{00}^0\left[2+\sqrt{\frac{g_1}{g_2}}+\sqrt{\frac{g_2}{g_1}}\right]\frac{1}{4\sqrt{1-g_1g_2}} \\ V_{00}^0 &= \frac{1}{2}\pi L w_{0s}^2 = \frac{L^2\lambda}{2} \qquad \left.\begin{aligned} & g_1g_2\to 1 \\ \text{or } & g_1g_2\to 0 \text{ but } g_1\neq g_2 \end{aligned}\right\} V_{00}\uparrow \end{aligned}\right\}(2.8.8)$$

Mode volume (high-order) $$\frac{V_{mn}}{V_{00}}=\frac{V_{mn}^0}{V_{00}^0}=\sqrt{(2m+1)(2n+1)}$$

For square spherical cavity $$V_{mn}^0=\frac{1}{2}L\pi w_{ms}w_{ns}=\sqrt{(2m+1)(2n+1)}V_{00}^0$$

3. Equivalent phase surface (wave-front)

$$R(z)=\left|z+\frac{f^2}{z}\right|=\left|f\left(\frac{z}{f}+\frac{f}{z}\right)\right|$$

$$f^2=\frac{L(R_1-L)(R_2-L)(R_1+R_2-L)}{\left[(L-R_1)+(L-R_2)\right]^2}$$

(2.8.4)

4. Far-field divergence angle (base mode)

$$\theta=\lim_{z\to\infty}\frac{2w(z)}{z}$$

$$\theta_{1/e^2}=\lim_{z\to\infty}\frac{2\sqrt{\frac{L\lambda}{2\pi}\left(1+\frac{z^2}{f^2}\right)}}{z}=2\sqrt{\frac{\lambda}{f\pi}}$$

$$=2\sqrt{\frac{\lambda}{\pi L}}\left\{\frac{\left[g_1+g_2-2g_1g_2\right]^2}{g_1g_2(1-g_1g_2)}\right\}^{1/4}$$

5. Resonant frequency

Square stable spherical cavity

$$
\left.
\begin{aligned}
\Phi_{mn}(r,z) &= -\left[k\left(z+\frac{r^2}{2R}\right)+kf-(m+n+1)\left(\frac{\pi}{2}-arctg\frac{f-z}{f+z}\right)\right] \\
\Phi_{mn}(r,z) &= -\left[k\left(z+\frac{r^2}{2R}\right)+kf-(m+2n+1)\left(\frac{\pi}{2}-arctg\frac{f-z}{f+z}\right)\right]
\end{aligned}
\right\} \quad (2.8.10)
$$

Circular stable spherical cavity

$$2\Delta\Phi_{mn}(r,z)=2\left[\Phi_{mn}(0,z_2)-\Phi_{mn}(0,z_1)\right]=-2q\pi \qquad (2.8.11)$$

insert z_1, z_2 and f

Stable square spherical cavity

$$\nu_{mnq}=\frac{c}{2\eta L}\left[q+\frac{1}{\pi}(m+n+1)arc\cos\sqrt{g_1g_2}\right] \qquad (2.8.12)$$

Stable circular spherical cavity

$$\nu_{mnq}=\frac{c}{2\eta L}\left[q+\frac{1}{\pi}(m+2n+1)arc\cos\sqrt{g_1g_2}\right] \qquad (2.8.13)$$

6. Diffraction loss

The mode loss is determined by N

Fresnel number $$N=\frac{a^2}{L\lambda}=\frac{a^2}{\pi w_{0s}^2} \Leftarrow w_{0s}=\sqrt{\frac{L\lambda}{\pi}}$$

General stable spherical cavity = = confocal cavity

So diffraction loss follow same rules, same traveling wave field

when $$\frac{a_i^2}{\pi w_{s_i}^2}=\frac{a_0^2}{\pi w_{0s}^2}$$ they have same δ_{mn}

N_{ef_i} Effective Fresnel number of stable spherical cavity

insert into (2.8.6) and (2.8.7)

$$\left.\begin{aligned} N_{ef1}&=\frac{a_1^2}{\pi w_{s_1}^2}=\frac{a_1^2}{|R_1|\lambda}\sqrt{\frac{(R_1-L)(R_1+R_2-L)}{L(R_2-L)}}=\frac{a_1^2}{L\lambda}\sqrt{\frac{g_1}{g_2}(1-g_1g_2)} \\ N_{ef2}&=\frac{a_2^2}{\pi w_{s_2}^2}=\frac{a_2^2}{|R_2|\lambda}\sqrt{\frac{(R_2-L)(R_1+R_2-L)}{L(R_1-L)}}=\frac{a_2^2}{L\lambda}\sqrt{\frac{g_2}{g_1}(1-g_1g_2)} \end{aligned}\right\} \quad (2.8.18)$$

According to the relationship between N and δ_{mn}, insert N to obtain $\delta_{mn}{}^1$ and $\delta_{mn}{}^2$